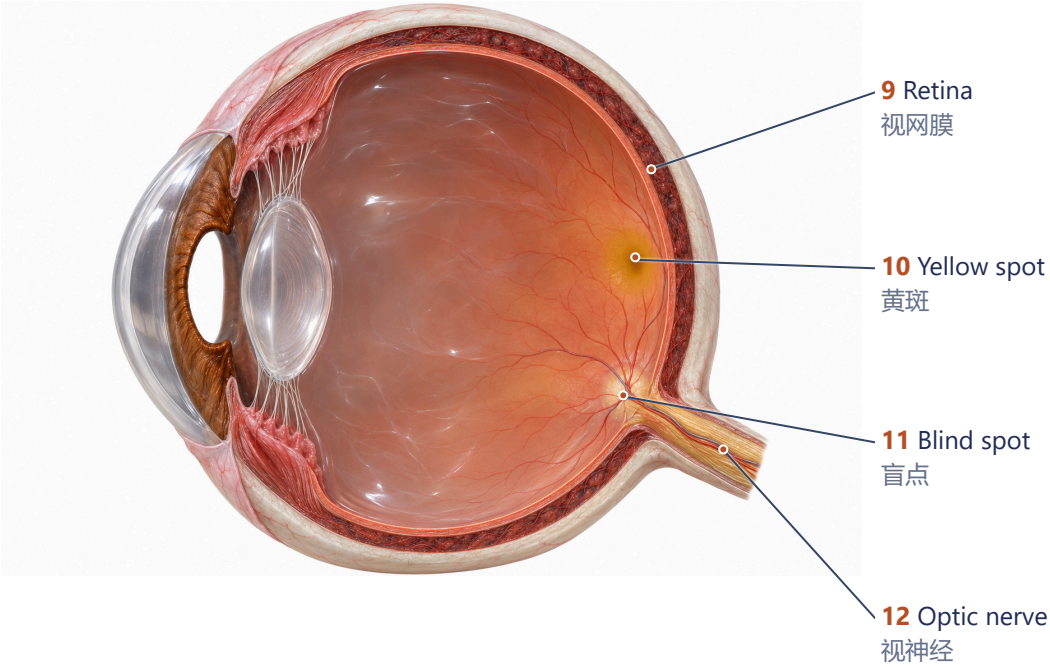
Eye: detecting light

眼睛：感光

Stimuli and Responses in Humans

人类的刺激与反应



Retina + optic nerve

视网膜与视神经

The retina produces nerve impulses. The _____ carries them to the brain.

视网膜产生神经冲动。_____把冲动传到脑。

Yellow spot + blind spot

黄斑与盲点

The yellow spot gives sharp vision. The blind spot has no photoreceptors.

黄斑提供清晰视觉。盲点没有感光细胞。

Rods

杆细胞

Rod cells help us see in _____ light. They do not distinguish colours.

杆细胞帮助我们在_____光下看见物体。它们不分辨颜色。

Cones

锥细胞

Cone cells support colour vision in bright light. Three types respond best to different ranges of colour.

锥细胞在明亮环境下提供色觉。三种类型对不同色光范围最敏感。

Quick check

1. Why is a coloured shirt harder to identify in dim light?
为甚么在暗处较难辨认衣服的颜色?

A. Rods detect dim light but not colour.
杆细胞感应微光，但不分辨颜色。

B. The iris produces colours.
虹膜产生颜色。

C. The blind spot becomes the whole retina.
整个视网膜变成盲点。

D. Cones become more sensitive than rods in dim light.
在弱光下，视锥细胞比视杆细胞更敏感。

2. Which retinal cells detect colour?
哪种视网膜细胞感受颜色?

A. Rod cells
视杆细胞

B. Cone cells
视锥细胞

C. Nerve fibres
神经纤维

D. Iris muscles
虹膜肌肉

3. Why does light falling on the blind spot produce no image signal?
为什么光落在盲点上时不产生视觉讯号?

A. The lens is opaque.
晶状体不透明。

B. The pupil closes.
瞳孔闭合。

C. It has no photoreceptors.
那里没有感光细胞。

D. The cornea absorbs all light.
角膜吸收全部光线。

4. Which retinal region gives the sharpest vision?
视网膜哪个区域提供最清晰的视觉?

A. Yellow spot
黄斑

B. Blind spot
盲点

C. Optic nerve
视神经

D. Choroid
脉络膜